

SUMAVARSHITHA KAMATAM

Senior DevOps & AI Engineer | Platform Engineering | LLM Agents & Automation
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PROFESSIONAL SUMMARY

DevOps & AI Engineer with 7+ years of experience building and scaling enterprise AI/GenAI systems, agentic workflows, and cloud-native DevOps platforms. Experienced in designing production-grade LLM agents, multi-agent orchestration, RAG, tool calling, MCP, LangGraph, GenSQL, and Google ADK/Gemini for intelligent automation, data insights, and engineering workflows. Currently leading release validation for SAP S/4HANA at SAP Labs, executing 400+ CI/CD pipelines/week at ~99% success rate, while driving Python-based automation across Jenkins, GitHub Actions, Kubernetes, Docker, FastAPI, and GCP. Proven track record of mentoring engineers, delivering scalable automation, and translating GenAI capabilities into measurable enterprise impact.

CORE TECHNICAL SKILLS

Languages	Python (primary), Java, Groovy, SQL, Bash
AI / GenAI	LLM Agents (single & multi-agent), Tool Calling, RAG, Prompt Engineering, MCP Patterns, Agentic Orchestration, LangChain, Google ADK, Antigravity, LangGraph, GenSQL, Bigquery
Automation	CI/CD Pipeline Design & Optimization, Jenkins, GitHub Actions, Ansible, Groovy scripting
Backend	FastAPI, Spring Boot, REST Microservices, Async & Event-Driven Workflows
DevOps / Infra	Docker, Kubernetes (deployments, scaling, orchestration), Linux, Helm, Infrastructure-as-Code (Terraform)
Cloud	AWS (ML Specialty Certified), SAP BTP (Cloud Foundry), Google Cloud Platform (GCP)
Observability	Grafana, Centralized Logging & Alerting, Tracing
Static Analysis	SonarQube, Nexus IQ, Security-focused Automation Pipelines
Databases	SAP HANA, PostgreSQL
Tools	JIRA, ServiceNow, Git

PROFESSIONAL EXPERIENCE

Senior DevOps & AI Engineer | SAP Labs India

Sep 2023 – Present

AI Agents & Intelligent Automation

- Designed and deployed a production-grade AI Failure Analysis Agent (tool-calling + RAG) integrated with ServiceNow MCP, enabling real-time incident enrichment, context-aware root cause identification, and auto-generated resolution suggestions, reducing duplicate incidents by ~40% and debugging effort by ~40%.
- Built LLM-powered agentic systems with single-agent and multi-agent reasoning patterns for security and engineering workflows, including automated incident classification, failure grouping, and template generation.
- Applied systematic prompt engineering to ensure reliable, auditable, and deterministic automation outputs.
- Developed GenAI UI system that converts natural language prompts into dynamic dashboards (charts, comparisons, trend analysis).
- Built an agentic data-insights platform using LangGraph + GenSQL, enabling API-based database connectivity, autonomous schema/data-context discovery, meaningful metric generation, and Human-in-the-Loop feedback for iterative refinement of AI-generated insights.

Python Automation & Backend Engineering

- Architected fully automated CI/CD orchestration system (Docker + Jenkins + Groovy) executing 400+ pipelines/week at ~99% success rate, reducing manual provisioning by ~90% and improving execution time by 30–40%.
- Built Data Insights Platform (Spring Boot + React + FastAPI) for enterprise analytics and AI-based cost optimization, led end-to-end development using Cline (AI-assisted development) and designed lightweight integration APIs (FastAPI) for AI-driven automation workflows.
- Led and mentored a team of engineers on AI agent development, Python automation best practices, and GenAI-driven tooling, driving adoption across the platform team.
- Implemented async and batch workflow patterns for large-scale pipeline automation, balancing latency and throughput requirements.
- Designed cost optimization framework, reducing unused infrastructure costs by ~20–25%.

Systems Integration & DevOps / Platform Engineering

- Built centralized orchestration platform for triggering pipelines, modifying runtime parameters, and monitoring infrastructure health in real-time via DCT Console Application (DCA) using Spring Boot, SAP UI5, HANA, and BTP.
- Developed TLV System Provisioning Platform enabling on-demand S/4HANA + CBC environments, improving environment availability and deployment velocity.
- Implemented and configured end-to-end SAP BTP setup (Subaccounts, XSUAA, Destinations, HANA, Cloud Connector) ensuring high availability and fault tolerance.

Jenkins & CI/CD Engineer | NTT DATA Business Solutions (SAP Client)

Oct 2021 – Aug 2023

- Developed and optimized Groovy-based CI/CD pipelines for large-scale S/4HANA release validation across distributed environments, automating end-to-end test scenarios (Initial, Upgrade, Update, Regression), reducing manual effort by ~60%.
- Designed modular, reusable pipeline frameworks, accelerating new team onboarding by ~40% and improving pipeline maintainability.
- Provisioned CBC & S/4HANA environments using Docker images and Kubernetes orchestration, including deployment configuration and activation across environments.
- Built monitoring & reporting systems, improving failure visibility and debugging efficiency; integrated external APIs/tools to enhance automation workflows.
- Optimized pipeline execution via parallel processing, improving throughput performance by ~25%.
- Collaborated cross-functionally with release, QA, and platform teams to resolve blockers and improve delivery timelines.

DevOps Engineer | Wipro Technologies (BT Client)

May 2020 – Sep 2021

- Designed and implemented scalable CI/CD pipelines (Jenkins, Maven) for enterprise applications, achieving ~95% build success rate.
- Built Grafana dashboards integrated with centralized logging & alerting systems, reducing incident detection time by ~30% and enabling proactive failure recovery.
- Automated deployments using Ansible & Docker, reducing manual setup effort by ~70%; introduced versioning and rollback strategies, reducing deployment failures by ~25%.
- Applied security best practices, compliance standards, and environment standardization across DevOps workflows.

DevOps Engineer | Wipro Technologies (HPSE Client)

Jul 2019 – May 2020

- Built end-to-end CI/CD pipelines for Spring Boot + React applications, accelerating release cycles and reducing manual intervention.
- Integrated SonarQube & Nexus IQ for code quality gate enforcement and artifact security scanning; standardized Git workflows for improved deployment reliability.
- Automated build, test, and deployment workflows using scripting and pipeline tooling.

KEY PROJECTS

- **AI Failure Analysis Agent (RAG + Tool Calling + MCP):** Production-grade agentic system analyzing CI/CD failures, mapping to historical incidents via vector retrieval, and auto-generating resolutions. Integrated ServiceNow MCP for live service data enrichment and context-aware suggestions. Reduced debugging effort by ~40%.
- **Multi-Environment GitHub Actions Deployment Pipeline:** Fully automated CI/CD workflow for microservices on Linux-based runners, including Docker build/push, automated testing, environment-specific deployment gates (dev/test/prod). Reduced release cycle time by ~35% and eliminated all manual deployment steps.

• DevOps Orchestration Platform

End-to-end orchestration platform for pipeline control, environment provisioning, and infrastructure health monitoring. Improved deployment efficiency and observability across distributed systems at enterprise scale.

• Security Automation Pipeline (Program Analysis)

Integrated static analysis tools (SonarQube, Nexus IQ) into CI/CD pipelines for automated vulnerability scanning, artifact security management, and dependency analysis; contributed to security-focused automation workflows.

- **GenAI Data Insights Platform:** Spring Boot + React + FastAPI platform enabling natural language-to-dashboard conversion with AI-driven cost analytics and trend visualization for enterprise stakeholders.
- **Agentic AI Career Guidance Platform:** Designed and deployed a cloud-native multi-agent system on Google Cloud Run using Google ADK, Gemini, BigQuery, Streamlit, and FastAPI; orchestrated agents for resume parsing, GitHub analysis, skill normalization, career-gap assessment, learning recommendations, and mock interview generation, enriched with BigQuery analytics and live job-market data to deliver personalized, role-specific career roadmaps and portfolio projects.
- **GenAI Data Insights & Metric Generation Platform:** Built an agentic data-insights platform using LangGraph and GenSQL that connects to databases through a secure API, autonomously discovers schemas, tables, columns, and data context, and generates meaningful application metrics and insights using natural-language data reasoning; implemented human-in-the-loop feedback

workflows allowing users to evaluate suggested metrics and trigger agent-driven refinement and regeneration.

ACHIEVEMENTS & CERTIFICATIONS

- AWS Machine Learning Specialty Certified
- Top 4 – SAP Invent to Customers AI Hackathon (RAG-based solution)
- Process Master Award for automation innovation and delivery excellence at SAP
- Promoted at SAP Labs for high-impact contributions to platform engineering and AI systems
- Conducted internal tech sessions on AI Agents, GenAI systems, and agentic automation best practices

EDUCATION

B.Tech – Electronics & Communication Engineering

Annamacharya Institute of Technology and Sciences

CGPA: 8.2